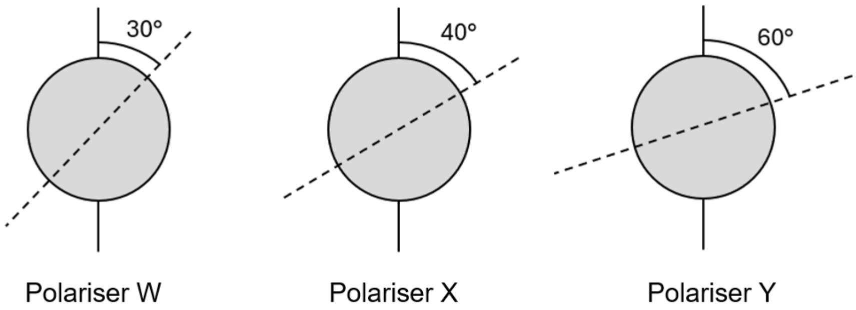
17 The diagram below shows three polarisers W, X and Y.

A beam of unpolarised light of intensity I_0 is incident normally onto the surface of polariser W, which then passes through polariser X and eventually emerges from polariser Y.

The axis of polarisation of each polariser is indicated by a dashed line.



What is the intensity of the light that emerges from polariser Y?

A $0.11\,I_0$ **B** $0.32\,I_0$ **C** $0.43\,I_0$ **D** $0.64\,I_0$

Ans: C

Using Malus' law, the intensity of transmitted light (I) through polarisers is proportional to $I_0 \cos^2\theta$, where I_0 is the intensity of transmitted through the first polariser, θ is the angle between two polarisers.

Unpolarized light loses exactly half of its intensity after it passes through polariser W no matter what the direction of polarising for polariser X is: 0.50 I_0.

The angle between polariser W and X is 10°, the transmitted intensity after X would be $0.50I_0 \times \cos^2(40° - 30°) = 0.48I_0$.

The angle between polariser X and Y is 20°, the transmitted intensity after Y would be $0.48I_0 \times \cos^2(60° - 40°) = 0.43I_0$.

18 A musical instrument is made using a long tube with a mouthpiece at one end. The other end is